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3D transient multiphase model for keyhole, vapor plume, and weld pool dynamics in laser welding including the ambient pressure effect

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Highlights

- We report a novel 3D transient multiphase model with ambient pressure effect.
- Without this effect, the average keyhole wall temperature is underestimated. Without this effect, the average speed of metallic vapor is overestimated.
- The ambient pressure is an essential physical factor in laser welding process.

Outline

Abstract

The physical process of deep penetration laser welding involves complex, self-consistent multiphase keyhole, metallic vapor plume, and weld pool dynamics. Currently, efforts are still needed to understand these multiphase dynamics. In this paper, a novel 3D transient multiphase model capable of describing a self-consistent keyhole, metallic vapor plume in the keyhole, and weld pool dynamics in deep penetration fiber laser welding is proposed. Major physical factors of the welding process, such as recoil pressure, surface tension, Marangoni shear stress, Fresnel absorptions mechanisms, heat transfer, and fluid flow in weld pool, keyhole free surface evolutions and solid–liquid–vapor three phase transformations are coupling considered. The effect of ambient pressure in laser welding is rigorously treated using an improved recoil pressure model. The predicated weld bead dimensions, transient keyhole instability, weld pool dynamics, and vapor plume dynamics are compared with experimental and literature results, and good agreements are obtained. The predicted results are investigated by not considering the effects of the ambient pressure. It is found that by not considering the effects of ambient pressure, the average keyhole wall temperature is underestimated about 500K; besides, the average speed of metallic vapor will be significantly overestimated. The ambient pressure is an essential physical factor for a comprehensive understanding the dynamics of deep penetration laser welding.

Keywords

Multiphase model; Keyhole dynamics; Weld pool dynamics; Vapor plume dynamics; Ambient pressure effect

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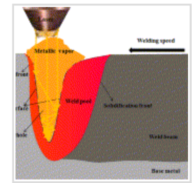
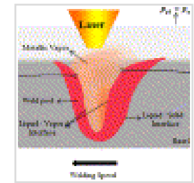
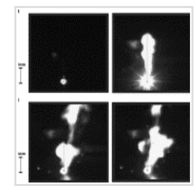
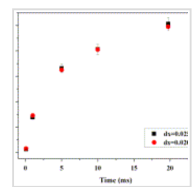
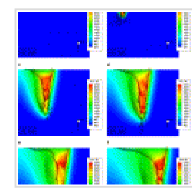
Figures (14)











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1. Introduction

Nowadays, deep penetration laser welding has been widely applied to aerospace, nuclear power, automotive, shipbuilding, and other important industries. In deep penetration laser welding, a slender vapor cavity, also called a keyhole which when filled with high temperature vapor plume or plasma, could be produced in an exothermic liquid weld pool due to the evaporation of the material. The transient keyhole, vapor plume, and weld pool dynamics are self-consistently correlated, and closely associated with the final quality of the laser weld joints. Mathematical modeling of the dynamics of the keyhole, vapor plume, and weld pool could be beneficial for the physical understanding and process optimization of deep penetration laser welding.

Many models have been proposed for understanding the physical process of deep penetration laser welding over the past 40 years. Earlier models [1], [2], [3], [4] mainly considered the heat transfer behaviors of the welding process, which were often used to predict the keyhole geometries or weld profiles. Possible fluid flow behaviors of a weld pool induced by thermal-capillary effect and vapor plume frictions were also studied based on calculated or observed fixed keyhole profiles [5], [6]. Their works have inspired researchers to study further the interactions between the keyhole, weld pool, and vapor plume in laser welding. However, these models mainly assumed that the keyhole has a quasi-steady profile, and cannot be used to simulate the evolution and oscillation of the keyhole from the beginning to a quasi-steady welding. Two-dimensional (2D) [7], [8] or three-dimensional (3D) [9] free surface based models, capable of describing the transient, self-consistent keyhole, weld pool dynamics, and even the formation process of porosity defects, were proposed for laser welding during the past decade. This work makes it possible to predict the transient evolution of the self-consistent keyhole and weld pool. Nevertheless, the plume dynamics inside the transient keyhole, which has important effects on the transient keyhole and weld pool dynamics [10], [11], cannot be predicted with these models. In recent years, several more prominent models were proposed to predict the self-consistent keyhole, weld pool, and vapor plume dynamics. The pioneering attempt was made by Ki et al. [12], [13] from the University of Michigan for CO_2 laser welding. In their work the main physical phenomena involved in laser welding process have been considered. The keyhole free surface was tracked using the Level Set method. A Symmetrically Coupled Gauss Seidel (SGGS) method was adopted to solve the model by using a powerful computer. Courtois et al. [14] have proposed a novel model based on Comsol Multiphysics software for keyhole, weld pool, and plume dynamics of Nd:YAG laser welding. In their model the energy transfer between the laser and keyhole interface was treated as a wave propagation problem by solving the Maxwell's equations, and the evaporation rate and the vaporization induced recoil pressure were included as localized source terms only at the keyhole interface. However, several important physical factors, such as Marangoni shear stress were neglected. Meanwhile, this model, as well as the models in [12], [13], [14] may achieve unphysical results near the interface since numerical smearing techniques are used to treat the discontinuous keyhole interface, according to Pang et al. [9]. Recently, Tan et al. [15] developed a multiphase model for keyhole, weld pool, and vapor plume dynamics for deep penetration Nd:YAG laser welding. The interactions between the molten pool, vapor plume, and assisting gas and its effects on the keyhole dynamics are investigated using their transient multiphase model. However, in their model the ambient pressure effect has not been taken into consideration.

Previous research has shown that the ambient pressure has significant effects on the penetration and defects in laser welding. For example, Katayama et al. [17] found that the state of the melt pool will become calmer when the ambient pressure decreases from atmospheric pressure to vacuum. Böerner et al. [18] demonstrated that the penetration depth will increase more than 110% in the low-speed laser welding process when the ambient pressure decreases from atmospheric pressure to vacuum. Amara et al. [6] showed that the ambient pressure has important effects on flow behaviors of the metallic vapor. The speed of the metallic vapor can increase to several thousand meters per second from less than 300 m/s, when the ambient pressure changes from 10 bar to 0.1 bar. Recently, Hirano et al. [19] found that the ambient pressure can reduce the evaporation rate when the temperature of evaporation surface is lower than or around the boiling point. The aforementioned research demonstrates that the ambient pressure may have great effects on the keyhole, weld pool, and metallic vapor dynamics in the laser welding process. Unfortunately, none of the previous comprehensive laser welding models [11], [12], [13], [14], [15], [16] has considered this important physical factor.

In this paper, a novel 3D multiphase transient model rigorously incorporating the effects of ambient pressure is proposed and validated. The model can simultaneously simulate the evolution of the transient self-consistent keyhole, vapor plume, and weld pool in the deep penetration laser welding process. Major physical factors such as recoil pressure, surface tension, Marangoni shear stress, buoyancy force, multiple reflections Fresnel absorption, heat transfer, and fluid flow of weld pool, gas–solid–liquid multiphase transformations are included in the model. The predicted keyhole wall temperature and speed of metallic vapor using the models without the effect of ambient pressure are investigated by comparing with the corresponding results that include this effect. The proposed model is evaluated by comparing the simulation results with in-situ imaging experiments of laser welding process and literature results.

2. Mathematic modeling

Here the fiber laser welding (1.07 μm wavelength) is mainly considered, although the model developed here can easily be extended to other laser welding processes. To make the model tractable, we make several simplifications. The beam profile is assumed to be Gaussian. Besides, since there are theoretical difficulties in finding proper boundary conditions for compressible modeling of the vapor plume in the transient keyhole, the vapor plume is assumed to be an incompressible gas. Moreover, the possible friction effect of the vapor plume towards the keyhole is neglected in the model, following most of previous free surface models [12], [14], [15], [16]. Fig. 1 shows the schematic of the deep penetration laser welding process. The major nomenclature used in this paper is given in Table 1. Other symbols in governing equations are defined when they are first introduced.

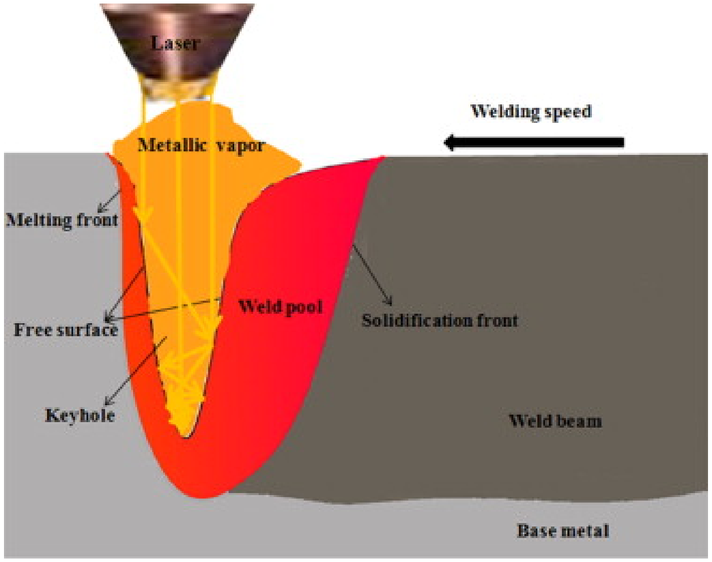

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Fig. 1. Schematic of deep penetration laser welding.

Table 1. Notation.

Symbol	Units	Meaning
Subscripts l, g, lb, gb	–	Molten liquid, vapor, boundary value of molten liquid and vapor
$\vec{U}$	m/s	Three dimensional velocity vector
ρ	kg/m^3	Density
$p, p_s, \overline{p}_s, p_{atm}$	N/m^2	Pressure, surface, average surface and ambient pressure
k	W/(mK)	Liquid heat conductivity
k_B	J/K	Boltzmann constant
$T, T_{ref}, T_s, T_l, T_v, T_\infty, T_{fs}, \overline{T}_{gb}$	K	Temperature of weld pool, reference, solidus, liquidus, boiling, ambient, keyhole wall and average boundary vapor temperature

Symbol	Units	Meaning
L_m, L_v	J/kg	Melting and evaporation latent
μ	$\mathrm{Ns/m^2}$	Dynamic viscosity
β	1/K	Liquid coefficient of thermal expansion
σ	N/m	Liquid surface tension
σ_s	$\mathrm{W/(m^2K^4)}$	Stefan–Boltzmann constant
$\frac{d\sigma}{dT}$	N/(mK)	Surface tension temperature coefficient
C_p	J/(kg K)	Liquid specific heat
ϕ	–	A time-dependent signed distance function
$\overrightarrow{g}$	$\mathrm{m/s^2}$	Gravitational vector
q	$\mathrm{W/m^2}$	The energy absorbed by multiple Fresnel absorption
h	$\mathrm{W/(m^2K)}$	The air convection coefficient
ε_r	–	The black body radiation coefficient
M_{atom}	kg/mol	The molar mass of metallic vapor
N_a	1/mol	The Avogadro number

2.1. Governing equations

2.1.1. Transient keyhole and weld pool dynamics

In the welding process the dynamics of the self-consistent keyhole and weld pool can be solved by our previous sharp interface (SI) model of Pang et al. [9]. The heat transfer and flow behaviors of an incompressible molten liquid in a weld pool can be described in [9]

$$\nabla \bullet \overrightarrow{U}_l = 0, \tag{1}$$

$$\rho_l \left(\frac{\partial \overrightarrow{U}_l}{\partial t} + (\overrightarrow{U}_l \bullet \nabla) \overrightarrow{U}_l \right) = \nabla \bullet (\mu_l \nabla \overrightarrow{U}_l) - \nabla p_l - \frac{\mu_l}{K} \overrightarrow{U}_l - \frac{C \rho_l}{\sqrt{K}} \left| \overrightarrow{U}_l \right| \overrightarrow{U}_l + \rho_l \overrightarrow{g} \beta (T - T_{ref}), \tag{2}$$

$$\rho_l C_p \left(\frac{\partial T}{\partial t} + (\overrightarrow{U}_l \bullet \nabla) T \right) = \nabla \bullet (k \nabla T), \tag{3}$$

where C is an inertial coefficient related to the liquid fraction f_l, $C = 0.13 f_l^{-3/2}$ [9]. K is the Carman–Kozeny coefficient of mixture model. The time-dependent keyhole free surface, i.e., the vapor plume–weld pool interface is tracked by the Level Set method [20]

$$\frac{\partial \phi}{\partial t} + \overrightarrow{U}_l \bullet \nabla \phi = 0, \tag{4}$$

2.1.2. Vapor plume dynamics

Here, the vapor plume or plasma in the keyhole in deep penetration laser welding is mainly assumed to be influenced by recoil pressure, ambient pressure, and assisting gas. Previous experiments have demonstrated that in fiber laser welding the ionization degree of a high temperature vapor plume is very small [21], as a consequence the gaseous phase in the keyhole can be assumed to be completely full with metallic vapor plume. Based on above considerations, the dynamics of the vapor plume in the keyhole can also be described by incompressible Navier–Stokes equations, as follows:

$$\nabla \bullet \overrightarrow{U}_g = 0, \tag{5}$$

$$\rho_g \left(\frac{\partial \overrightarrow{U}_g}{\partial t} + (\overrightarrow{U}_g \bullet \nabla) \overrightarrow{U}_g \right) = \nabla \bullet (\mu_g \nabla \overrightarrow{U}_g) - \nabla p_g + \rho_g \overrightarrow{g}, \tag{6}$$

2.1.3. Surface pressure model

The previous recoil pressure model [22], which was widely adopted in theoretical modeling of laser welding [7], [8], [9], [10], [11], [12], [13], [14], [15], is independent of ambient pressure. However, recent research [19], [23] showed that the evaporation process is actually influenced by recoil pressure and ambient pressure. Therefore, an improved recoil pressure model dependent on ambient pressure, called a surface pressure model and developed by Pang et al. [23], is used to describe the evaporation process in laser welding. The surface pressure model can be described as follows [23]:

$$p_s\left(T_{fs}\right) = \begin{cases} p_{amb} & 0 \le T_{fs} < T_L \\ \frac{1+\beta_R}{2} p_0 \exp\left(\frac{\Delta H_v}{k_B T_v}\left(1 - \frac{T_v}{T_{fs}}\right)\right) & +\infty > T_{fs} \ge T_R \\ p_c(T_{fs}) & T_L \le T_{fs} < T_R \end{cases}, \tag{7}$$

where T_v is the boiling temperature under the atmospheric pressure p_0, β_R is the recombination rate which is set to a constant 0.2 in this paper, ΔH_v is the enthalpy of phase transition during vaporization, $p_c \ (T_{fs})\,(T_L \le T_{fs} < T_R)$ is a smooth cubic interpolation curve, which is set to make the temperature dependent ambient pressure curve smooth, T_L and T_R are the temperatures of the two tangent points between the smooth curve $p_c \ (T_{fs})$ with the ambient line and the recoil pressure curve. The parameters of the surface pressure model for laser welding of stainless steel under atmospheric pressure are referred to in [23].

2.2. Boundary conditions

In mathematic models for deep penetration laser welding, boundary conditions especially the ones at the discontinuous liquid–vapor interface are complex and important. Fig. 2 is the schematic of boundary conditions in deep penetration laser welding. S shown in view I and S' in view II of Fig. 2, are area elements of the molten liquid and metallic vapor at the keyhole free surface, respectively. $\vec{n}$ is the unit normal vector of the keyhole free surface, and $\vec{t}_1$ and $\vec{t}_2$ are the unit orthogonal tangent vectors perpendicular to the normal of the keyhole free surface. The boundary conditions for weld pool dynamics and metallic vapor flow in laser welding are described as follows.

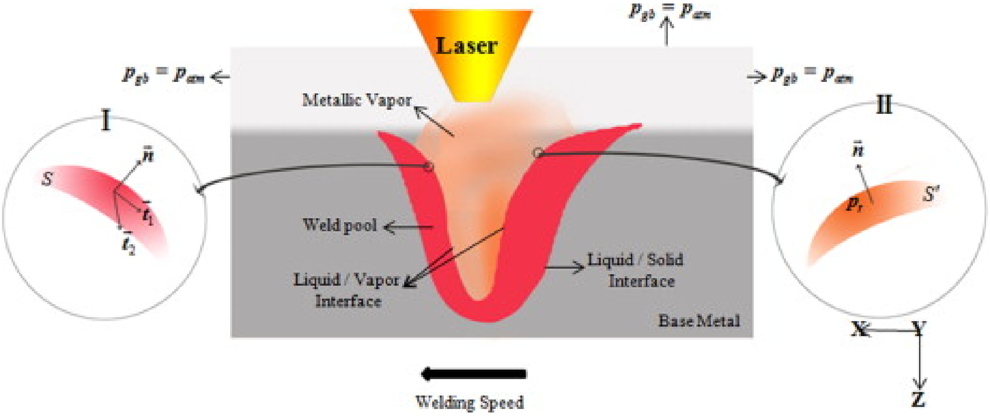

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Fig. 2. Boundary conditions in deep penetration laser welding.

2.2.1. Boundary conditions for weld pool dynamics

In the weld pool of laser welding, the surface pressure, surface tension, and hydrodynamic pressure are acting on S at the normal of the keyhole free surface, and the Marangoni shear stress and viscous stress are acting on S at the tangent of the keyhole free surface. The pressure and viscous stress boundary conditions of molten liquid at the keyhole free surface are expressed by the following equation, respectively [9]:

$$p_{lb} = p_s + \sigma\kappa + 2\mu_l \vec{n} \bullet \nabla \vec{U}_l \bullet \vec{n}, \tag{8}$$

$$\begin{aligned}\left(\mu_l \nabla \vec{U}_l\right)_{lb} = {} & \mu_l \left(\vec{n} \quad \vec{t}_1 \quad \vec{t}_2\right)\left(\vec{n} \quad \vec{0} \quad \vec{0}\right)^T \left(\nabla \vec{U}_l\right)\left(\vec{n} \quad \vec{0} \quad \vec{0}\right)\left(\vec{n} \quad \vec{t}_1 \quad \vec{t}_2\right)^T \\ & + \mu_l \left(\vec{n} \quad \vec{t}_1 \quad \vec{t}_2\right)\left(\vec{0} \quad \vec{t}_1 \quad \vec{t}_2\right)^T \left(\nabla \vec{U}_l\right) \\ & - \mu_l \left(\vec{n} \quad \vec{t}_1 \quad \vec{t}_2\right)\left(\vec{n} \quad \vec{0} \quad \vec{0}\right)^T \left(\nabla \vec{U}_l\right)^T \left(\vec{0} \quad \vec{t}_1 \quad \vec{t}_2\right)\left(\vec{n} \quad \vec{t}_1 \quad \vec{t}_2\right)^T \\ & + \left(\vec{n} \quad \vec{t}_1 \quad \vec{t}_2\right)\begin{pmatrix} 0 & \nabla_s \sigma \bullet \vec{t}_1 & \nabla_s \sigma \bullet \vec{t}_2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\left(\vec{n} \quad \vec{t}_1 \quad \vec{t}_2\right)^T,\end{aligned} \tag{9}$$

where p_{lb} is the pressure boundary value of molten liquid at the keyhole free surface interface, κ the curvature of the keyhole free surface, $\left(\mu_l \nabla \vec{U}_l\right)_{lb}$ is the viscous stress boundary value of molten liquid at the keyhole free surface interface, $\nabla_s = \left(\mathrm{I} - \vec{n}\vec{n}\right)\nabla$ is the surface gradient operator, $\nabla_s \sigma$ is the Marangoni shear stress.

For the molten liquid on the keyhole wall, by considering the multiple Fresnel absorption, heat convection, thermal radiation, and evaporation, the thermal boundary conditions are described as follows [9]:

$$k \frac{\partial T}{\partial \vec{n}} = q - h(T - T_\infty) - \varepsilon_r \sigma_s (T^4 - T_\infty^4) - \rho_l V_{evp} L_v, \tag{10}$$

where V_{evp} is the keyhole interface recession speed due to evaporation [12], [13]. To the molten liquid on other surfaces, the thermal boundary conditions are expressed as follows [9]:

$$k\frac{\partial T}{\partial \vec{n}} = -h(T - T_\infty) - \varepsilon_r \sigma_s (T^4 - T_\infty^4), \tag{11}$$

2.2.2. Boundary conditions for metallic vapor flow

The metallic vapor is mainly driven by surface pressure p_s. Therefore, the pressure boundary condition for metallic vapor on the keyhole free surface is described as follows:

$$p_{gb} = p_s, \tag{12}$$

The outlet pressure boundary condition for metallic vapor flow is easily described as follows:

$$p_{gb} = p_{atm}, \tag{13}$$

Ignoring the effects of the Knudsen layer, the temperature of metallic vapor on the keyhole free surface is assumed to be the same as the one on the keyhole wall. Therefore, the thermal boundary condition of metallic vapor on the keyhole surface is described as follows:

$$T_{gb} = T_{fs}, \tag{14}$$

Assuming that the vapor behaves as a perfect gas, the density ρ_g of the metallic vapor can be approximated as follows:

$$\rho_g = \left(\frac{M_{atom}}{N_a k_B}\right) \frac{\overline{p}_s}{\overline{T}_{gb}}, \tag{15}$$

3. Numerical implementation

The developed 3D multiphase model was solved with high order finite difference methods with an in-house parallelized code. For the Level Set Eq. (4), a third order TVD Runge–Kutta scheme and a fifth order Weighted Essentially Non-Oscillatory (WENO) scheme were used for the discretization of the transient term and convective term, respectively. The Eikonal equation used to re-initialize the Level Set value at each time step [9] was solved with a highly efficient fast sweeping method [24], which only takes O(N) time where N is the grid number. For the fluid flow equations of the molten liquid in the weld pool (Eqs. (1), (2), (3)) and the vapor plume in the keyhole (Eqs. (5), (6)), the transient term was discretized with a first order forward difference scheme, the convective term with a fifth order WENO scheme and the diffusion term with a second order central difference scheme.

A staggered grid based projection method [25] was adopted to solve the Navier–stokes (NS) equations for the molten liquid and vapor plume. Taking the NS equations for the molten liquid as an example, a tentative velocity field, $\overrightarrow{U}_l^*$, was first calculated without considering the pressure gradient term,

$$\rho_l{}^n \frac{\overrightarrow{U}_l^* - \overrightarrow{U}_l^n}{\Delta t} + \widetilde{C}(\overrightarrow{U}_l^n) = \widetilde{D}(\mu_l{}^n \overrightarrow{U}_l^n) + \widetilde{K}(\overrightarrow{U}_l^n) + \widetilde{F}^n \tag{16}$$

where $\widetilde{C}(\bullet)$, $\widetilde{D}(\bullet)$, $\widetilde{K}(\bullet)$ and $\widetilde{F}^n$ are respectively represent the discretization operators of the convective term, diffusive term, Darcy term, and buoyancy term. Second, taking the pressure gradient into account, one can obtain

$$\rho_l{}^n \frac{\overrightarrow{U}_l^{n+1} - \overrightarrow{U}_l^*}{\Delta t} = -\nabla p_l{}^{n+1} \tag{17}$$

where $\overrightarrow{U}_l^{n+1}$ is the velocity at the new time step, $p_l{}^{n+1}$ is the pressure at the new time step. Let us take the divergence of the two sides of Eq. (17), then one can obtain a pressure Poisson equation for molten liquid of weld pool,

$$\nabla^2 p_l{}^{n+1} = \frac{\rho_l{}^n}{\Delta t} \nabla \bullet \overrightarrow{U}^* \tag{18}$$

A central difference scheme was used to discretize Eq. (18) by considering the Dirichlet pressure boundary conditions, and a parallel Preconditioned Conjugate Gradient (PCG)

[26] method was used to solve the systems of the linear equations of pressure. After the pressure field was calculated, it was then substituted into Eq. (17) to obtain the accurate velocity field of the molten pool at the new time step. For solving the NS equations of the vapor plume, a similar procedure was used in the present study.

The energy equation Eq. (3) was discretized using a semi-implicit method: an explicit scheme for convective term and implicit scheme for diffusion term. The semi-discretization of Eq. (3) is as follows:

$$\rho_l{}^n C_p{}^n \frac{T^{n+1}-T^n}{\Delta t} + C'(T^n) = D'(k^{n+1}T^{n+1}) \tag{19}$$

where $C'(\bullet)$ and $D'(\bullet)$ are the discretization operators of the convective term and diffusive term, respectively. During the discretization process, the convective heat transfer boundary conditions (Eqs. (10), (11)) were incorporated, and the latent heat of melting and solidification was also included using a temperature back compensation method. A parallel PCG method was also used for solving the resulted linear equations of temperature. The possible Fresnel absorption of the keyhole wall was solved using a robust ray tracking method [27].

The calculation steps of the present model are shown in Fig. 3, as follows:

1) Calculate the absorbed energy of the keyhole free surface using ray tracing method.

2) Solve the energy equation Eq. (3) and obtain the temperature field of workpiece.

3) Solve the NS equations Eqs. (1), (2) to obtain the fluid flow of weld pool with respect to the boundary condition Eqns. (8), (9), and solve the NS equations Eqns. (5), (6) by considering the boundary conditions equation Eqns. (12), (13).

4) Update Level Set value to obtain the keyhole free surface at the new time step.

5) Go back to step 1) and start the next step simulation.

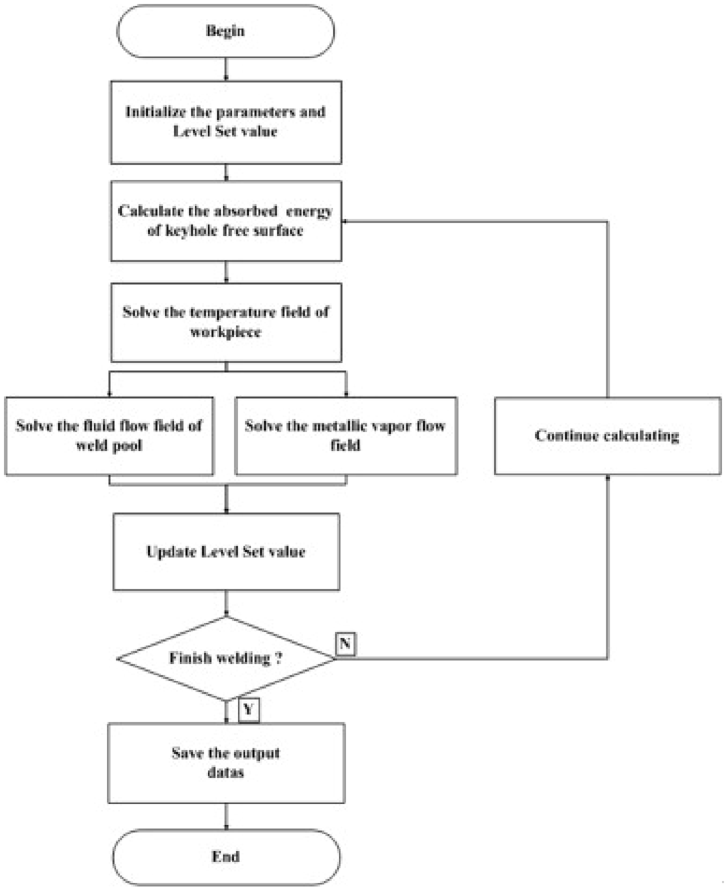

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Fig. 3. Solution flow chart of the multiphase mathematical model.

4. Experimental method

In this paper, deep penetration laser welding experiments of 3 mm thick 304 stainless steel were carried out with a continuous wave (CW) fiber laser (IPG YLR–4000, maximum power: 4 kW). The laser beam (1.07 μm wavelength), which is delivered by a fiber system and focused on the specimen surface, has a spot radius around 0.2 mm at the focal position. The chemical composition of 304 stainless steel is shown in Table 2. Before welding the specimen was carefully cleaned with acetone to eliminate possible oil stain and oxide film. During laser welding the welding process was monitored by a high speed CMOS camera (Photonfocus, Switzerland) in an argon shielding gas environment. A high speed photograph speed, 6000 frame per second (fps), was used in imaging to capture the detailed plume dynamics. Although the 3D multiphase model developed here can easily include the effect of the assisting gas, in our simulations this effect is neglected to reduce the computational cost. As a consequence, a very weak gas flux rate, 0.1 L/min, was used in the experiments. After welding, the welds were cut, ground, polished, and etched. Then, the weld dimensions were measured under an optical microscope (OM).

Table 2. Chemical composition of 304 stainless steel.

C	Si	Mn	P	S	Ni	Cr	Fe
0.07	0.46	0.78	0.032	0.006	8.10	18.32	Balance

Since it is difficult to observe the vapor plume inside the keyhole, only the vapor plume outside the keyhole is imaged. It can be observed that in laser welding, the vapor plume outside the keyhole is usually in dynamic oscillation. Hence, it is possible to use the height variations of the vapor plume image to estimate the ejecting speed of the vapor plume from the keyhole opening, called ES hereafter, in vertical direction. The ES can be estimated to be the division between the height variation of two adjacent images and the time interval, since the value of time interval, 1.0/6000=0.167 ms, is less than the oscillation period of the keyhole and metallic vapor plume [21]. In laser welding, the vapor plume may be broken into several parts. In such a case, the height measurement was only made on the white part that connects to the keyhole opening, as illustrated in Fig. 4.

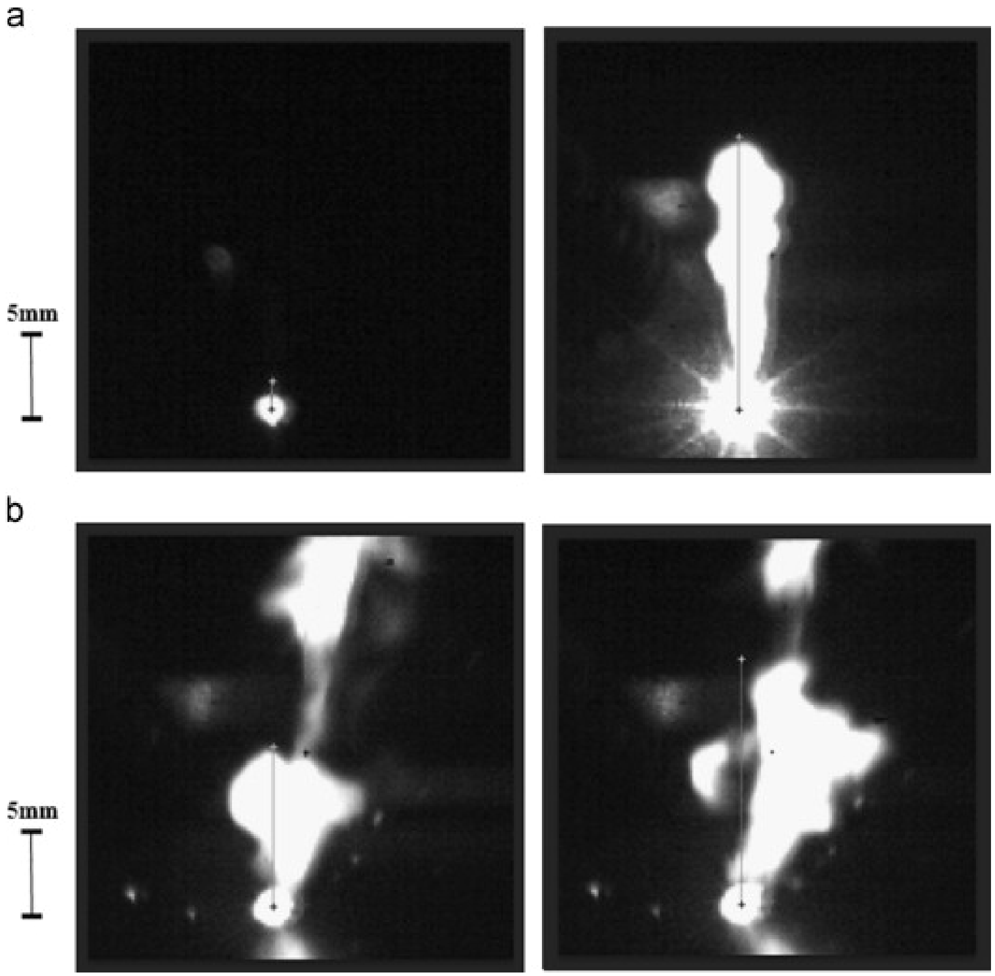

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Fig. 4. High-speed cameral images of vapor plume in laser welding (P=2 kW): (a) unseparated metallic vapor (time: t+0, t+0.167 ms); (b) separated metallic vapor (time: t+5.544, t+5.711 ms).

5. Results and discussion

To compare with experimental and literature results, simulations of fiber laser welding processes described in Section 4 are performed using the proposed model. The physical parameters used in the simulations are shown in Table 3. The temperature varying thermal parameters, such as heat conductivity, viscosity, density, and specific heat capacity, which are determined using the commercially available thermal parameter calculation software JMatPro, are averaged and assumed to be constants in the welding process. It is noted that the time step used in the simulations is very small (typically around 1.0×10^{-8} s). To make the calculation tractable, a small part of the specimens, sized in 3 mm×1.5 mm×3 mm in the welding, cross-section and penetration directions respectively, is used in the simulations.

Table 3. Physical parameters used in the simulations.

Property	Symbol	Value
Liquid density (kg/m^3)	ρ_l	7200
Liquid specific heat (J/(kgK))	C_p	760
Liquid heat conductivity (W/(mK))	k	35
Solidus temperature (K)	T_s	1697
Liquidus temperature (K)	T_l	1727
Boiling temperature (K)	T_v	3100
Melting latent (J/kg)	L_m	6.0×10^4
Evaporation latent (J/kg)	L_v	6.52×10^6
Liquid surface tension (N/m)	σ	1.0
Liquid dynamic viscosity (Ns/m^2)	μ_l	0.006
Gas dynamic viscosity (Ns/m^2)	μ_g	1.38×10^{-7}
Liquid coefficient of thermal expansion (1/K)	β	1.95×10^{-5}

Property	Symbol	Value
Surface tension temperature coefficient (N/(mK))	$\frac{d\sigma}{dT}$	-0.43×10^{-4}

5.1. Grid refinement study

To validate the numerical convergence of the simulation code, a grid refinement study is performed for the process with 2 kW power and 3 m/min welding speed, by setting the step at 0.025 mm and 0.02 mm, respectively. Fig. 5 shows the calculated time dependent penetration depth evolutions varying with grid step in deep penetration laser welding. It can be seen that the penetration depths with different grid steps are nearly the same, which demonstrates the numerical convergence of the simulations.

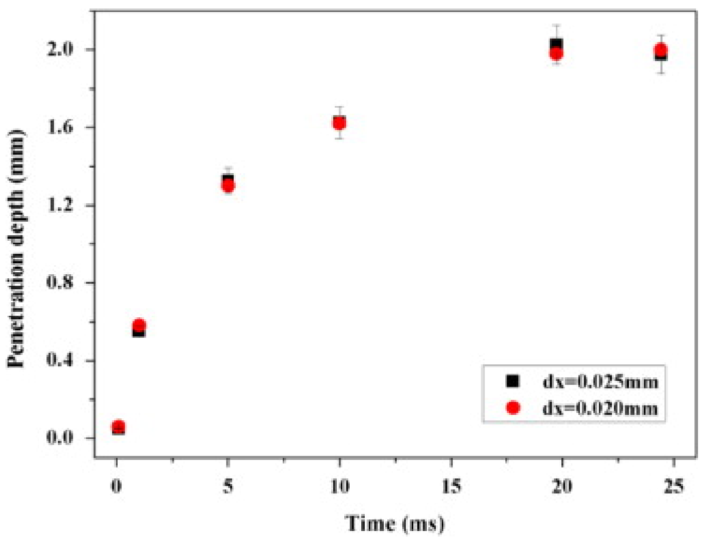

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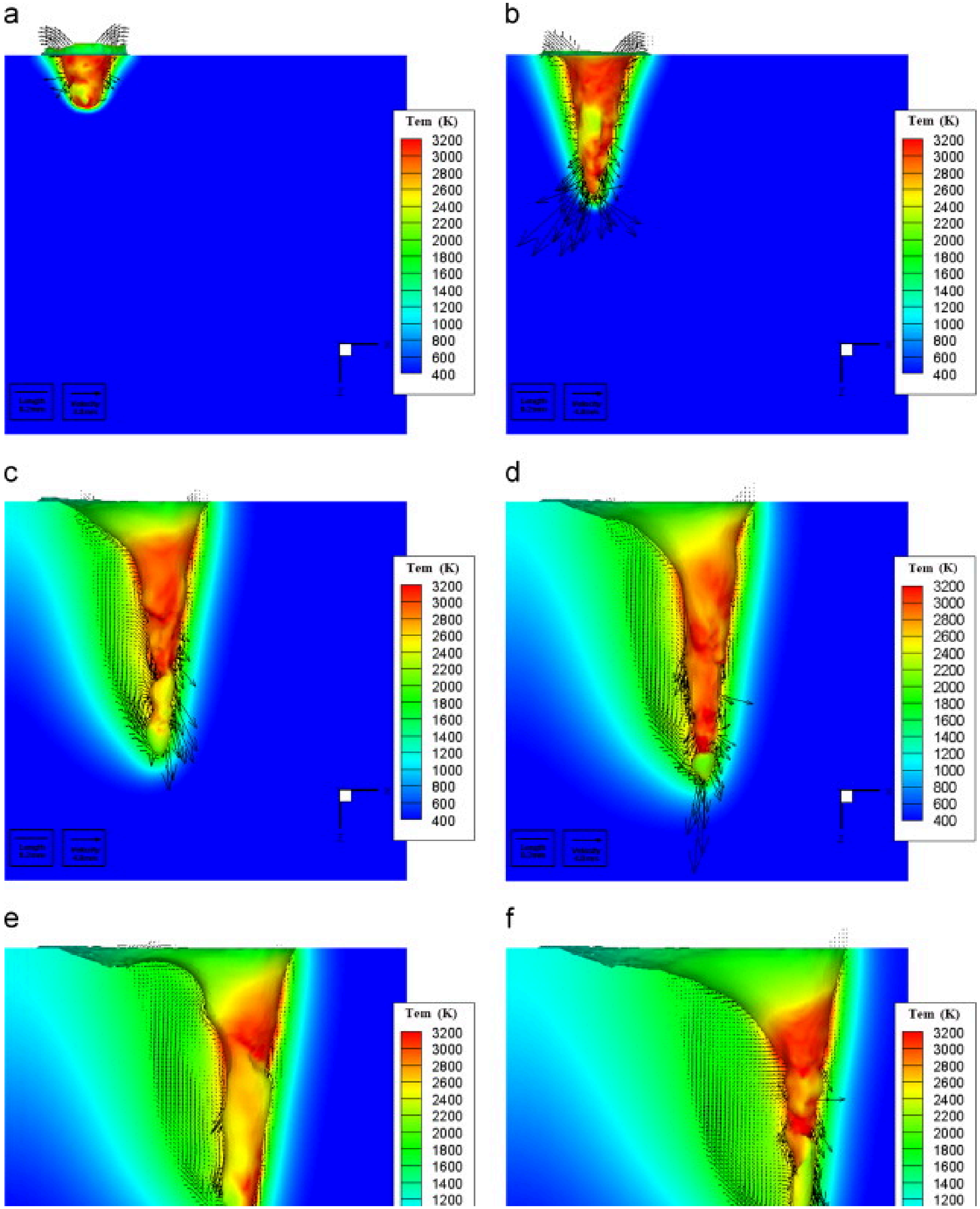
Fig. 5. Evolution of penetration depth with different grid steps.

5.2. Simulation results of the transient keyhole, weld pool and plume dynamics

Here the predicted keyhole, weld pool, and plume dynamics for laser welding with 2.0kW power and 3m/min welding speed are investigated. Fig. 6 shows the 3D transient keyhole profiles and weld pool dynamics. The dark arrow in Fig. 6 represents the velocity of the weld pool, and the color represents the temperature. It can be learned that the keyhole profile is highly unstable and many irregular humps can be found on the wall. These phenomena agree well with the previous imaging results of the keyhole, such as Matsunawa et al. [28], Katayama et al. [29], [30], Abt et al. [31] and Kaplan et al. [32]. Moreover, the average temperature of the keyhole wall is very high, approximately 2900K, that is also in line with the long believed assumption that the keyhole wall temperature is around the boiling point of the material [5], [33], [34], [35]. Moreover, there are very vigorous fluid flows (the maximum magnitude can be greater than 20m/s which is consistent with the recent reported experiments [36]) in the weld pool. Near the bottom part of front keyhole wall and behind the rear keyhole wall, there are interesting downward fluid motions (Fig. 6(d)–(f)). These phenomena are also closely correlated with X-Ray imaging results of weld pool dynamics in fiber laser welding of Katayama et al. [30].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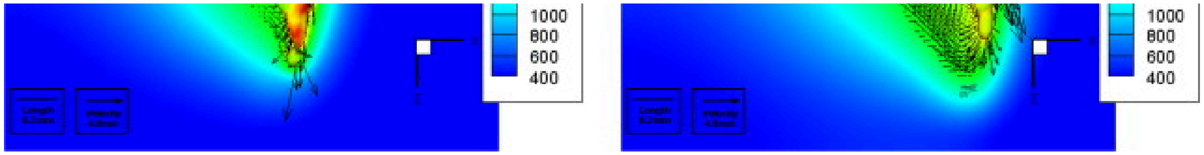

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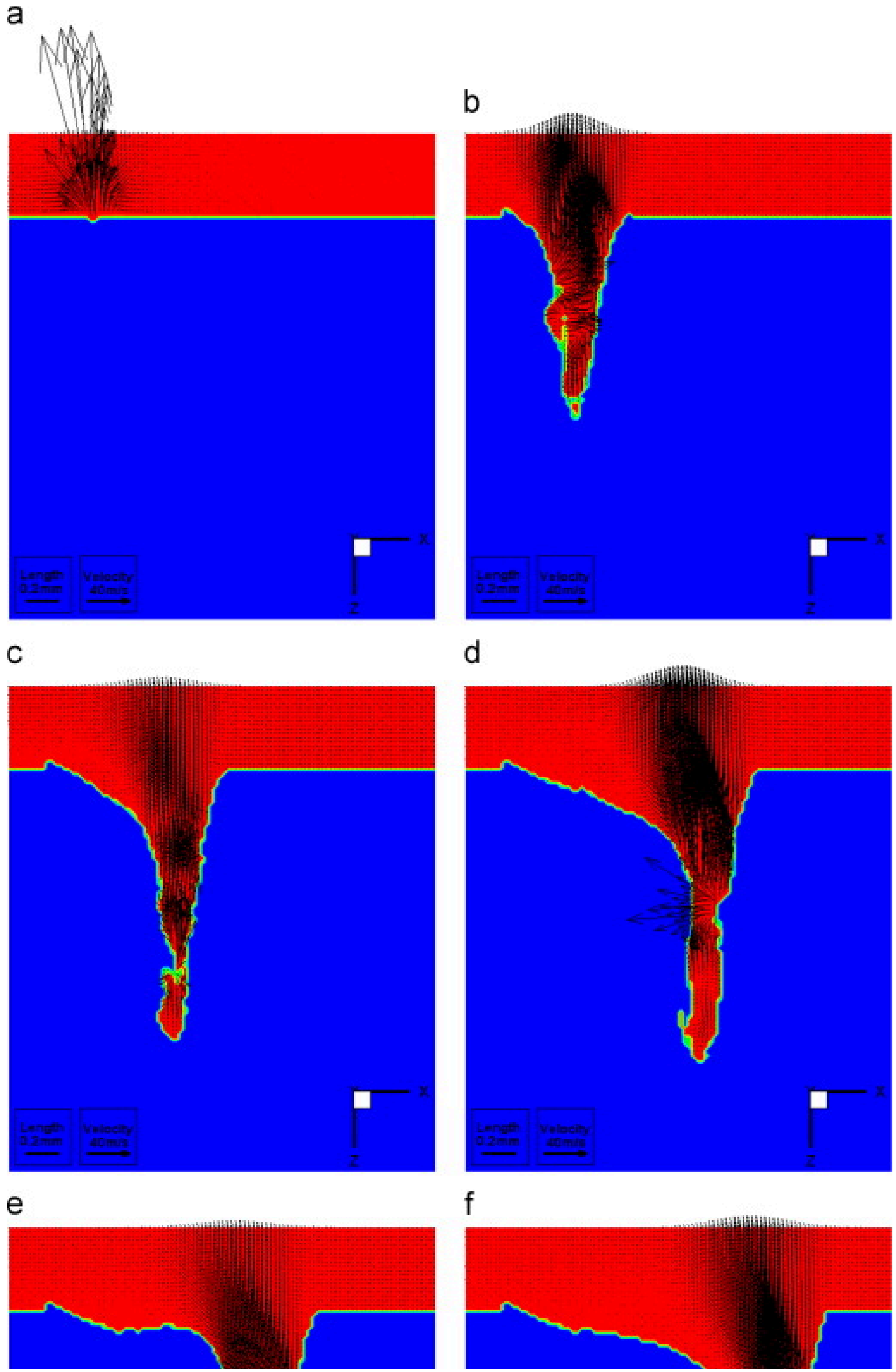
Fig. 6. Calculated temperature and velocity fields in weld pool and keyhole in laser welding (P=2 kW, V=3 m/min): (a) 0.493 ms; (b) 1.927 ms; (c) 10.222 ms; (d) 15.762 ms; (e) 20.855 ms; (f) 27.076 ms (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.).

Fig. 7 illustrates the time varied velocity distribution of metallic vapor in the transient keyhole (longitudinal section view). Fig. 8(a) shows the three dimensional view of vapor plume velocity and Fig. 8(b) shows the pressure (to make the calculation more convenient, the atmospheric pressure is taken as the zero level of pressure) distribution in the keyhole at 20.855 ms. It can be seen in Fig. 7, Fig. 8 that there are highly transient and vigorous vapor flows in the keyhole in deep penetration laser welding. From the pressure plot we see that the average relative pressure difference inside the keyhole is around 4500 Pa. It is noted that the vapor plume dynamics is mainly triggered by recoil pressure. Moreover, the free surface should have enough recoil pressure to withstand the surface tension to open the keyhole. Noting that the radius of the keyhole is around 0.2 mm, and keeping in mind that the surface tension coefficient σ is 1.0, it can be easily estimated that the vapor plume pressure inside the keyhole is around 5000 Pa. These data are very close to the present calculation result, around 4500 Pa.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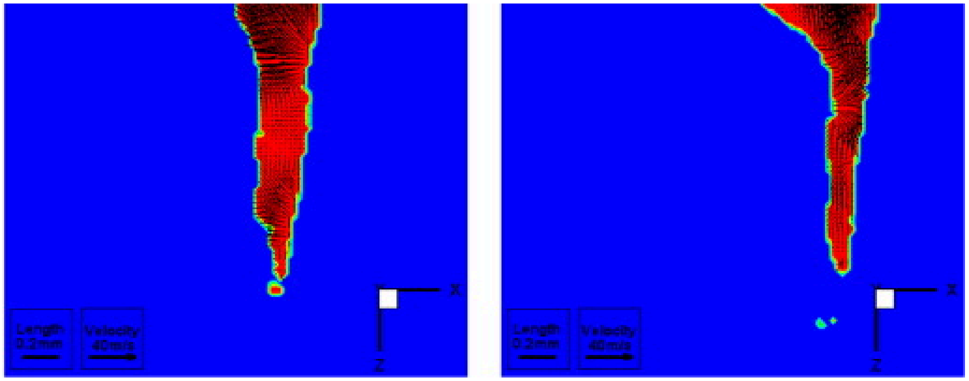

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Fig. 7. Longitudinal section views of calculated vapor velocity distributions in laser welding (P=2.0 kW, V=3 m/min): (a) 0.033 ms; (b) 3.754 ms; (c) 10.222 ms; (d) 18.715 ms; (e) 20.855 ms; (f) 27.129 m.

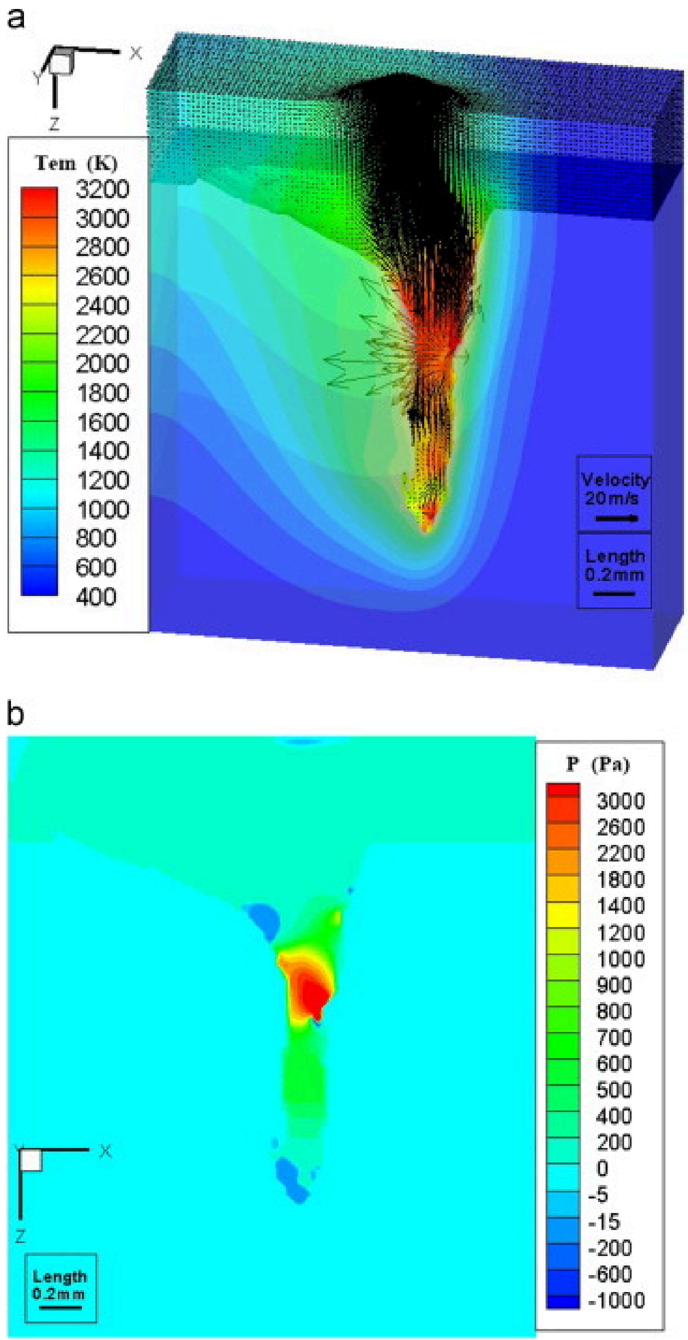

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Fig. 8. Calculated vapor velocity (3D view) and pressure at 20.855 ms (P=2.0 kW, V=3 m/min): (a) 3D velocity; (b) pressure distributions.

The aforementioned results demonstrate that the proposed 3D transient multiphase model is reasonable for describing keyhole profiles, weld pool dynamics, and plume pressure in deep penetration laser welding. In a later section, it will be shown that the calculated vapor velocity and weld pool dimensions also agree well with the experimental results.

5.3. Comparing with experimental results

Fig. 9 shows the comparison of weld dimensions between experimental and simulation in two welding processes (1.5 kW or 2.0 kW, V=3 m/min). The comparisons of fusion depth and middle fusion width are shown as Fig. 9(a) and (b) respectively. From the comparisons, it is clear that the predicted weld dimensions are in good agreement with experiments.

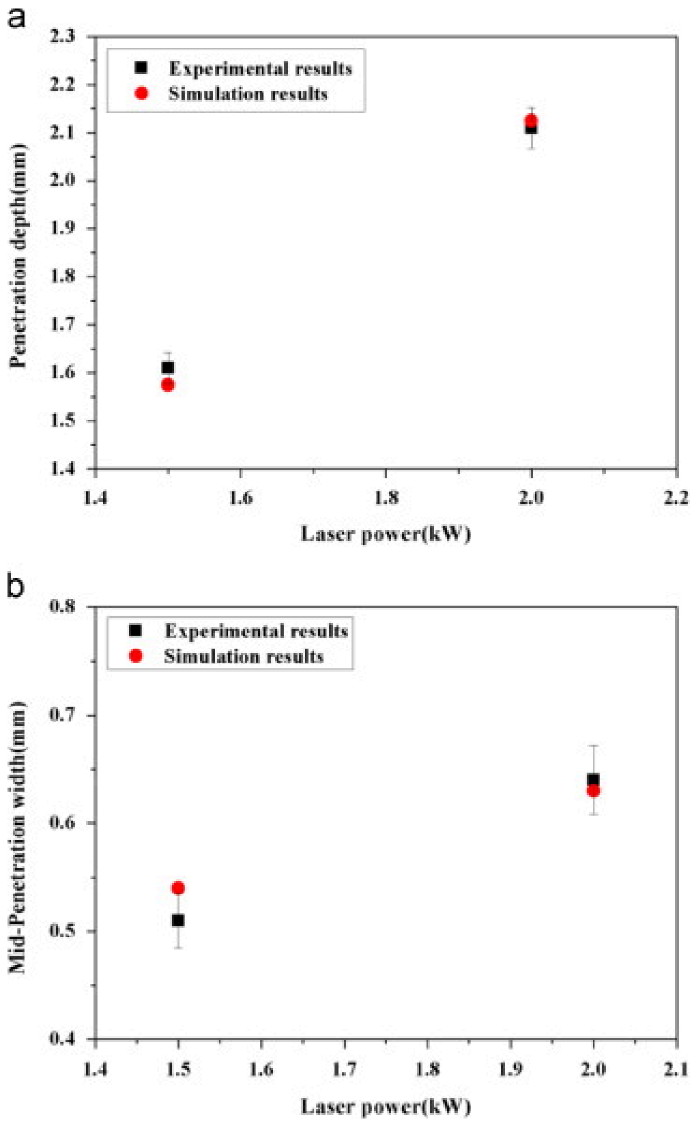

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Fig. 9. Weld dimension comparisons between experimental and simulation results: (a) fusion depth; (b) middle fusion width.

Fig. 10 shows the comparison of the ejection speed, ES, of the vapor plume at the keyhole opening in the *Z*-axis direction between experimental and simulation results. From the results shown in Fig. 10, it can be learned that from the initial time of welding up to 8 ms, ES rapidly rises up (90 m/s for 1.5 kW and 160 m/s for 2 kW) and then decreases within 1 ms. Hereafter ES will apparently oscillate, which indicates the keyhole is in dynamic oscillation. It can be seen that the predicted ES exhibits the same changing tendency. Besides, it can be also seen that the predicted average ES in high power welding process (2 kW) is slightly larger than that in low power (1.5 kW) case. From Fig. 10, it is noted that the ES is in the order of tens of meter per second, which seems to be relatively small. However, the maximum velocity of the vapor plume during laser welding at atmospheric pressure can reach several hundred meters per second (as shown in Fig. 11), that is consistent with previous theoretical calculations.

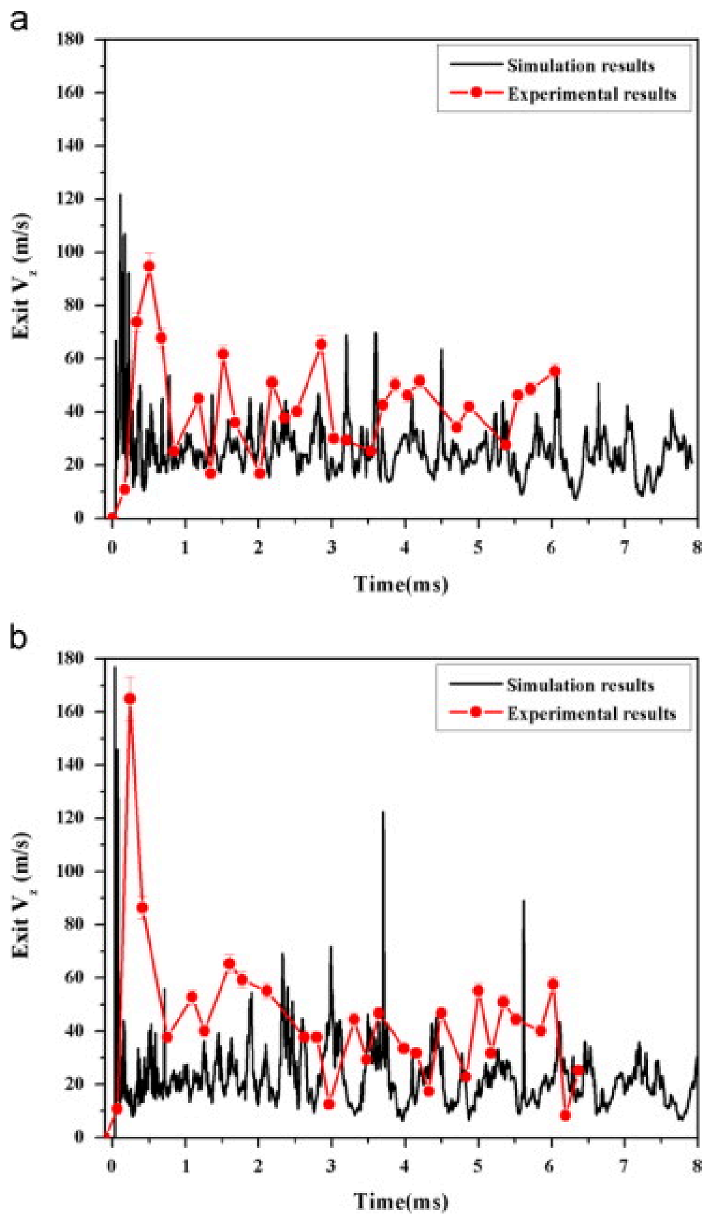

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Fig. 10. ES comparisons between experimental and simulation results: (a) P=1.5 kW, V=3 m/min; (b) P=2.0 kW, V=3 m/min.

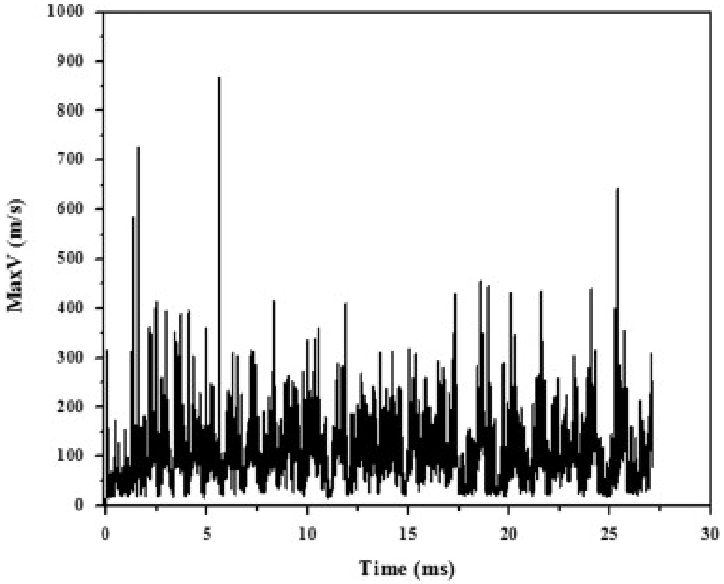

Fig. 11. Evolution of maximum velocity of vapor plume (P=2.0 kW, V=3 m/min).

The above results show the proposed model can be used to predict the weld dimensions and vapor plume velocity, which demonstrates the validity of the model again.

5.4. Discussion

Our results demonstrate that based on the developed model with an ambient pressure effect, the self-consistent keyhole evolutions, vapor plume, and weld pool dynamics can be well predicted. Below, some possible problems of the predicted keyhole temperature and vapor speed by not considering the effect of ambient pressure will be discussed.

Fig. 12 shows the comparison of simulated keyhole wall temperature by using the proposed model with (shown in Fig. 12(a)) and without (shown in Fig. 12(b)) the effect of ambient pressure. By considering the effects of atmospheric pressure, most of the keyhole wall temperature is around 2900 K or even higher. The highest one, which often occurs on the top of the humps because of being directly illuminated by the laser beam, can reach

3400K. So, the average of the keyhole wall temperature is around the boiling point of the material (3100K) or even higher with the atmospheric pressure effect. However, without this effect, the average of the keyhole wall temperature is around 2400K, which is much lower than the boiling point of material (3100K) at the atmospheric pressure. Recent experiments have shown that the ambient pressure can confine metallic vapor, when the temperature of evaporation surface is lower than or around the boiling point [19]. Therefore, by not considering the influence of ambient pressure, the simulated keyhole wall temperature may be remarkably underestimated (around 500K as shown in Fig. 12).

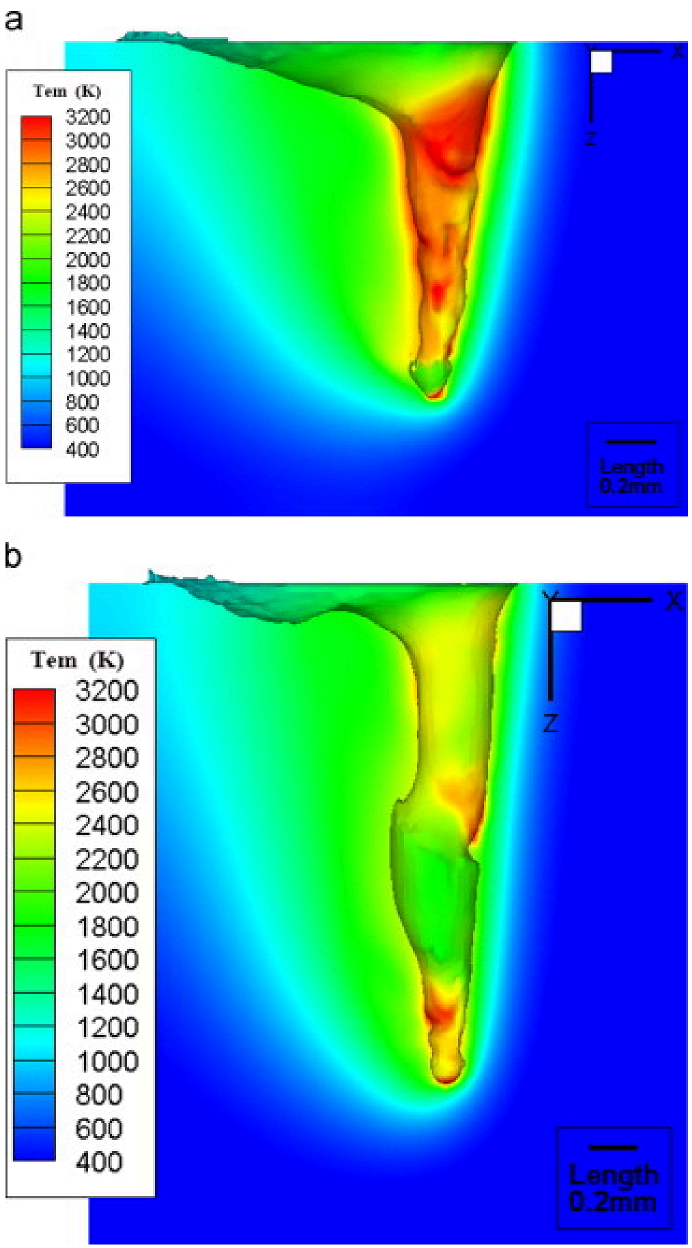

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Fig. 12. Comparison of the predicted keyhole wall temperature at 20.855 ms in laser welding at the atmospheric pressure (P=1.5 kW, V=3 m/min) with (a) or without ambient

pressure effect (b).

Fig. 13, Fig. 14 show the comparison of the velocity distribution and maximum velocity of metallic vapor by using the proposed model with and without the effects of ambient pressure. By considering the effect of ambient pressure, the predicted average speed of metallic vapor is between several tens of meters per second to several hundreds of meters per second (Fig. 7 and Fig. 13(a)). However, by not considering this effect, the average vapor speed can easily be greater than 300m/s (Fig. 13(b)). This means that there may be approximately one order of magnitude difference between the predicted metallic vapor speed with and without the ambient pressure effect.

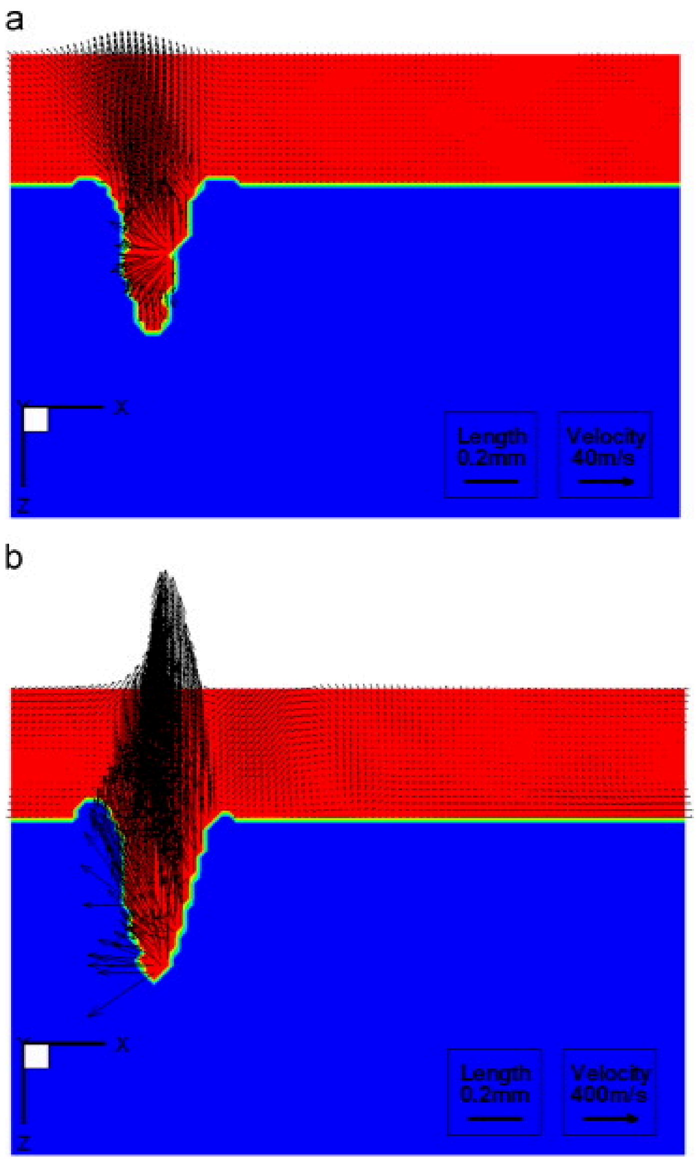

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Fig. 13. Comparison of the predicted velocity distributions of metallic vapor at 1.237 ms in laser welding (P=1.5 kW, V=3 m/min) with (a) or without ambient pressure effect (b).

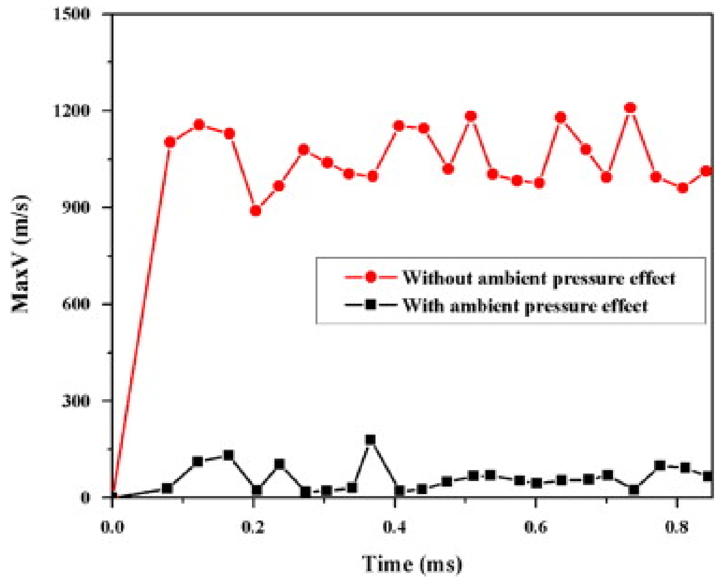

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Fig. 14. Comparison of the predicted maximum velocity of metallic vapor in laser welding (*P*=1.5 kW, *V*=3 m/min) with (a) or without ambient pressure effect (b).

In the present study, although the peak velocity of vapor plume can be higher than 300 m/s, the predicted average vapor plume speeds at the keyhole opening along vertical direction are only around 35 m/s (laser power: 1.5 kW, welding speed: 3 m/min) and 40 m/s (laser power: 2 kW, welding speed: 3 m/min), respectively, that are consistent with our experimental results (Fig. 10). It is noted that the vapor speed depends on the process parameters. Previously, several pioneering research studies [12], [13], [14], [15], [16], [37] have investigated the vapor plume dynamics and shown significant high speed vapor behaviors in the keyhole under different process parameters. For example, Amara et al. [37] predicted that the average ejection vapor velocity from the keyhole is around 300 m/s (laser power: 10 kW, welding speed: 20 cm/s). These data are rather large as compared to our present results. This difference can result from the facts below. First, the penetration depth (4 mm) is more than 2 times higher than those in the present study (1.61 and 2.11 mm), which indicates that there is much more energy absorbed on the keyhole wall.

Second, the vapor flow is modeled by enforcing a Dirichlet velocity boundary condition at a static keyhole wall boundary, which is quite different from our present study as well as the real physical context. Their way of modeling could be prone to produce higher speed vapor dynamics, as the boundary layer effect and the recession effect of the dynamic keyhole have been neglected. More importantly, the ambient pressure effect has been neglected, so that the predicated vapor plume dynamics is actually a rough approximation of that in vacuum laser welding, as discussed below. Recent experiments [21] indicated that the average ejection vapor speed from the keyhole opening during fiber laser welding under atmospheric pressure is only around 33–44 m/s (the laser power: 3 kW, welding speed: 1 m/min) which is consistent with our theoretical and experimental data, demonstrating the fact again that the average ejection velocity of vapor at keyhole opening is relatively slow at moderate laser power conditions.

When the ambient pressure effect is not considered, the density of vapor plume inside the keyhole could be very small as the welding process is actually a kind of vacuum laser welding. In the present theoretical modeling, the density of vapor without the ambient pressure effect is set to be 1/100 of atmospheric pressure, that is, 0.01 kg/m^3. As shown in Fig. 13(b) and Fig. 14, it can be found that the average vapor speed in this case is actually around 300 m/s at least, and the peak velocity can be easily greater than 1000 m/s, which is in good agreement with previous theoretical studies of vapor speed during laser material interaction under vacuum conditions [22], [38].

We note that previous theoretical multiphase or single phase free surface modeling of laser welding [12], [13], [14], [15], [16], [37] did not include the effect of ambient pressure. Their results, for example [13], showed that the average vapor plume speed is around 100 m/s during laser welding. Besides, during their calculations, the vapor plume density is usually artificially enlarged, which is physically inaccurate. For example, in the study of Ki et al. [13], the density is set to be 10 kg/m^3, which is approximately one order of magnitude higher than the actual density during welding under atmospheric pressure. If they put a more accurate vapor plume density, e.g. 1.0 kg/m^3, into the theoretical modeling, the predicted vapor velocity would be easily greater than 1000 m/s, which could be significantly greater than the present and previous experimental [21] values. Therefore, neglecting the ambient pressure effect, the theoretical predictions of vapor plume speed could be significantly overestimated.

Based on the discussions, therefore, the ambient pressure cannot be ignored in the comprehensive model for deep penetration laser welding process, if one needs to correctly predict the dynamics of self-consistent keyhole, vapor plume, and weld pool.

6. Conclusions

1. A novel 3D transient multiphase model with ambient pressure effect has been developed for deep penetration fiber laser welding. Transient keyhole instability, vapor plume dynamics, and weld pool dynamics can be quantitatively predicted with the model. Good agreements are obtained between simulation and experimental and literature results.

2. For the investigated process conditions, it is shown that with the ambient pressure effect, the average of predicted keyhole wall temperature is around 2900K or even higher, which is around the boiling point of the material (3100K). Besides, the average relative pressure difference inside the keyhole is around 4500Pa, and the average ejection speed of metallic vapor at the keyhole opening is of the order of tens of meters per second.

3. Without considering the ambient pressure effect, the average of keyhole wall temperature is around 2400K, which is at least 500K lower than the experiment results. Besides, the predicted speed of metallic vapor could be significantly overestimated.

Acknowledgment

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Simulation of melt pool behaviour during additive manufacturing: Underlying physics and progress

Cook P.S., Murphy A.B.

Additive Manufacturing • Volume 31 • 2020 • Article 100909

> *Citation Excerpt:*
>
> …On the other hand, Semak and Matsunawa [30] quote a private communication from Anisimov that the recoil coefficient does remain close to the minimal value for ‘practical values of the ambient pressure’. Eq. (18) has been used in simulations of keyhole laser welding with factors of 0.54 [31–34], 0.55 [35] and 0.6 [36], and with a factor of 0.54 in simulations of laser powder bed fusion [37–41]. To explain observed variations in laser penetration depth with the ambient pressure Pamb, Pang et al. [42] recommend that a minimum value of Pamb should be imposed on the recoil pressure obtained from eq. (18).…

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Research article

Dal M., Fabbro R.

Optics and Laser Technology • Volume 78 • 2016

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